

CS 506: Introduction to Quantum Computing

Measurement Postulate, Entanglement, and Projective Measurement

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1 Measurement (Observation) Postulate

Measurement destroys the state.

Question. In a multi-qubit state, can we observe an individual qubit, or a subset of the qubits?

Answer. It depends.

Take the two-qubit state

$$|\psi\rangle = \sqrt{\frac{1}{11}} |00\rangle + \sqrt{\frac{5}{11}} |01\rangle + \sqrt{\frac{2}{11}} |10\rangle + \sqrt{\frac{3}{11}} |11\rangle, \quad (1)$$

where in $|b_1b_2\rangle$ the left bit is the first qubit and the right bit is the second.

1.1 Standard (Computational) Basis

Observe both qubits in the standard basis:

$$\begin{aligned} |00\rangle \quad \text{with probability} \quad \left| \sqrt{\frac{1}{11}} \right|^2 &= \frac{1}{11}, & |01\rangle \quad \text{with probability} \quad \frac{5}{11}, \\ |10\rangle \quad \text{with probability} \quad \frac{2}{11}, & & |11\rangle \quad \text{with probability} \quad \frac{3}{11}. \end{aligned}$$

Can we observe one qubit without destroying the others?

1.2 Observing One Qubit

Group the terms of (1) by the first qubit and factor it out:

$$|\psi\rangle = |0\rangle \otimes \left(\sqrt{\frac{1}{11}} |0\rangle + \sqrt{\frac{5}{11}} |1\rangle \right) + |1\rangle \otimes \left(\sqrt{\frac{2}{11}} |0\rangle + \sqrt{\frac{3}{11}} |1\rangle \right).$$

The two groups have squared norms $\frac{1}{11} + \frac{5}{11} = \frac{6}{11}$ and $\frac{2}{11} + \frac{3}{11} = \frac{5}{11}$. Pulling these out leaves normalized states inside:

$$|\psi\rangle = \sqrt{\frac{6}{11}} |0\rangle \otimes \left(\sqrt{\frac{1}{6}} |0\rangle + \sqrt{\frac{5}{6}} |1\rangle \right) + \sqrt{\frac{5}{11}} |1\rangle \otimes \left(\sqrt{\frac{2}{5}} |0\rangle + \sqrt{\frac{3}{5}} |1\rangle \right). \quad (2)$$

Measuring only the first qubit, $|\psi\rangle$

$$\begin{aligned} \text{becomes } & |0\rangle \otimes \left(\sqrt{\frac{1}{6}} |0\rangle + \sqrt{\frac{5}{6}} |1\rangle \right) \quad \text{with probability } \frac{6}{11}, \\ \text{becomes } & |1\rangle \otimes \left(\sqrt{\frac{2}{5}} |0\rangle + \sqrt{\frac{3}{5}} |1\rangle \right) \quad \text{with probability } \frac{5}{11}. \end{aligned}$$

The first qubit collapses; the second is left in superposition, not destroyed.

2 Bell Basis and Entanglement

What about this? One of the four Bell basis states,

$$|\Phi_0\rangle = \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle, \quad (3)$$

cannot be represented like (2), i.e.

$$|\Phi_0\rangle \neq (a_0 |0\rangle + a_1 |1\rangle) \otimes (b_0 |0\rangle + b_1 |1\rangle)$$

for any a_0, a_1, b_0, b_1 satisfying $|a_0|^2 + |a_1|^2 = 1$ and $|b_0|^2 + |b_1|^2 = 1$. Expanding the product gives $a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle$, so matching (3) needs

$$a_0 b_0 = \frac{1}{\sqrt{2}}, \quad a_0 b_1 = 0, \quad a_1 b_0 = 0, \quad a_1 b_1 = \frac{1}{\sqrt{2}}.$$

Since $a_0 b_0 \neq 0$, both $a_0 \neq 0$ and $b_0 \neq 0$; then $a_1 b_0 = 0$ forces $a_1 = 0$ and $a_1 b_1 = 0$, a contradiction.

Here measuring one qubit also measures the other: outcome 0 leaves $|00\rangle$ and outcome 1 leaves $|11\rangle$, each with probability $\frac{1}{2}$. This is **quantum entanglement** — “spooky action at a distance” (Einstein). Tearing the qubits apart destroys them; too much power is needed to keep them entangled.

3 Projective Measurement

So far we have discussed the **von Neumann measurement**. A more general measurement is the **projective measurement**, given by

$$\underbrace{\mathbb{I}}_{\text{identity}} = \underbrace{P_1 + P_2 + \cdots + P_n}_{\text{projectors}}, \quad P_i^2 = P_i \quad \text{for all } i. \quad (4)$$

Assume $P_i^\dagger = P_i$ for all i ; then they form a basis. (Projectors are Hermitian, not unitary.) From (4), multiplying on the left by P_j ,

$$P_j = P_j \sum_i P_i = P_j^2 + \sum_{i \neq j} P_j P_i = P_j + \sum_{i \neq j} P_j P_i \implies P_i P_j = 0 \quad \text{for } i \neq j.$$

So any $|\psi\rangle$ can be written in terms of $P_1, \dots, P_n$:

$$|\psi\rangle = \mathbb{I} |\psi\rangle = \sum_{i=1}^n P_i |\psi\rangle = p(1) |\psi_1\rangle + p(2) |\psi_2\rangle + \cdots + p(n) |\psi_n\rangle,$$

where $P_i |\psi\rangle = p(i) |\psi_i\rangle$ with $|\psi_i\rangle$ normalized. Closure gives

$$|p(1)|^2 + |p(2)|^2 + \cdots + |p(n)|^2 = 1,$$

and the state becomes $|\psi_i\rangle$ with probability $|p(i)|^2$.

4 Measurement in Dirac Notation

Say $|b_1\rangle, |b_2\rangle, \dots, |b_n\rangle$ are a basis of a qubit system, with $\langle b_i | b_j \rangle = \delta_{ij}$, and

$$|\psi\rangle = b_1 |b_1\rangle + b_2 |b_2\rangle + \cdots + b_n |b_n\rangle.$$

Observing $|\psi\rangle$: it becomes $|b_i\rangle$ with probability $|b_i|^2$, where

$$|b_i|^2 = b_i^\dagger b_i$$

is the amplitude of a complex number. By orthonormality,

$$\langle b_i | \psi \rangle = b_i.$$

Instead of $|\psi\rangle$ we take the transpose (conjugate transpose) of $|\psi\rangle$:

$$\langle \psi | = b_1^\dagger \langle b_1 | + b_2^\dagger \langle b_2 | + \cdots + b_n^\dagger \langle b_n |,$$

so $\langle \psi | b_i \rangle = b_i^\dagger$. Therefore

$$\text{probability of } |b_i\rangle = |b_i|^2 = b_i^\dagger b_i = \langle \psi | b_i \rangle \langle b_i | \psi \rangle \quad \text{in Dirac notation.}$$

4.1 Parity

Let $|b_1\rangle, \dots, |b_n\rangle$ be the standard basis of an n -qubit system. Then

$$\mathbb{I} = P_0 + P_1, \quad P_0 = \sum_{\text{parity}(b_i)=0} |b_i\rangle \langle b_i|, \quad P_1 = \sum_{\text{parity}(b_i)=1} |b_i\rangle \langle b_i|,$$

where $\text{parity}(b_i)$ is the XOR of the bits of b_i . This measurement reveals only the parity, not the full string. For two qubits,

$$P_0 = |00\rangle \langle 00| + |11\rangle \langle 11|, \quad P_1 = |01\rangle \langle 01| + |10\rangle \langle 10|.$$

Next: Universal Gates